

DATA PREPROCESSING AND BUSINESS QUESTIONS

Strategic Product Placement Analysis

Data Preprocessing Overview

Data preprocessing prepares the retail dataset for accurate and meaningful Tableau analysis. The dataset is reviewed to ensure that the fields required for product placement and sales analysis are understandable, consistent, and suitable for visualization.

1. DATASET REVIEW	Review the retail dataset and identify the fields related to product category, product position, sales volume, consumer demographics, foot traffic, product price, competitor price, promotion, and season.
2. FIELD VALIDATION	Check field names and verify that each attribute represents the intended retail measure or dimension.
3. MISSING VALUE CHECK	Review the data for missing or incomplete values that could affect comparisons and visual results.
4. DATA TYPE CHECK	Confirm that sales volume, foot traffic, and price fields are treated as numerical values and category-related fields are suitable for grouping.
5. CONSISTENCY CHECK	Check categorical values such as product position, promotion status, season, and demographics for consistent naming.
6. TABLEAU READINESS	Load the reviewed dataset into Tableau and verify dimensions, measures, aggregations, and field usability before building charts.

BUSINESS QUESTIONS

Questions Used to Guide Tableau Analysis

No.	Business Question	Analysis Focus
1	What is the average sales volume across different product categories?	Product Category and Average Sales Volume
2	How does product position affect average sales volume?	Product Position and Average Sales Volume
3	How does sales volume vary across consumer demographic groups?	Consumer Demographics and Sales Volume
4	What is the relationship between foot traffic and average sales volume?	Foot Traffic and Average Sales Volume
5	How does product price compare with competitor price?	Product Price and Competitor Price
6	How do promotional activities influence sales performance?	Promotion and Sales Volume
7	How does sales performance vary across seasons?	Season and Sales Volume
8	Which combination of retail factors reveals useful product placement patterns?	Category, Position, Traffic, Pricing, Promotion, Season, and Demographics

Development Outcome

The preprocessing steps ensure that the retail dataset is suitable for Tableau analysis. The business questions provide a focused direction for chart development and help connect each visualization to a specific retail analytics need. These questions form the basis of the interactive dashboard and analytical story used in Strategic Product Placement Analysis.

Team Lead

bhavyagogeneni@gmail.com

Members

chilakanagapravallika@gmail.com

karramaniroshan@gmail.com

naaperusandeep333@gmail.com

kasturibhanu162@gmail.com